

SHETH LUJ AND SIR MV COLLEGE
Subject: Data Analysis with SAS / SPSS / R

Practical No: 12

Aim: Combining datasets vertically (concatenation) using rbind() (R).

Code:

```
# R Script: Vertical Concatenation using rbind() / bind_rows()
library(dplyr)
```

```
# --- 1. SETUP: Load Data ---
```

```
df <- read.csv("Cancer.dataset.csv") %>%
  select(id, diagnosis, radius_mean, texture_mean)
```

```
# --- 2. Create Two Separate Datasets ---
```

```
df_batch_1 <- head(df, 5)
```

```
df_batch_2 <- data.frame(
  id = c(999001, 999002),
  diagnosis = c("M", "B"),
  radius_mean = c(15.50, 12.30),
  texture_mean = c(20.10, 18.50)
)
```

```
print("--- Data Structure Check ---")
print(names(df_batch_1))
print(names(df_batch_2))
```

```
# 3. VERTICAL COMBINATION
```

```
combined_data <- bind_rows(df_batch_1, df_batch_2)
```

```
print("--- Combined Data Summary ---")
print(paste("Batch 1 rows:", nrow(df_batch_1)))
print(paste("Batch 2 rows:", nrow(df_batch_2)))
print(paste("Total Combined rows:", nrow(combined_data)))
```

```
print("--- Preview (Bottom rows show the new batch) ---")
print(tail(combined_data))
```

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Output:

The screenshot shows the RStudio interface with the following components:

- Source Editor:** Contains R code for loading data, creating two batches, and combining them.
- Console:** Displays the output of the R code, including data structure checks and a preview of the combined data.
- Environment:** Lists the objects created in the R session, such as `Cancer.dataset`, `df`, `df_batch_1`, `df_batch_2`, and `combined_data`.

```
> # R Script: Vertical Concatenation using rbind() / bind_rows()
> library(dplyr)
> # --- 1. SETUP: Load Data ---
> df <- read.csv("Cancer.dataset.csv") %>%
+   select(id, diagnosis, radius_mean, texture_mean)
> # --- 2. Create Two Separate Datasets ---
> df_batch_1 <- head(df, 5)
> df_batch_2 <- data.frame(
+   id = c(999001, 999002),
+   diagnosis = c("M", "B"),
+   radius_mean = c(15.50, 12.30),
+   texture_mean = c(20.10, 18.50)
+ )
> print("--- Data Structure Check ---")
[1] "--- Data Structure Check ---"
> print(names(df_batch_1))
[1] "id"      "diagnosis" "radius_mean" "texture_mean"
> print(names(df_batch_2))
[1] "id"      "diagnosis" "radius_mean" "texture_mean"
> # 3. VERTICAL COMBINATION
> combined_data <- bind_rows(df_batch_1, df_batch_2)
> print("--- Combined Data Summary ---")
[1] "--- Combined Data Summary ---"
> print(paste("Batch 1 rows:", nrow(df_batch_1)))
[1] "Batch 1 rows: 5"
> print(paste("Batch 2 rows:", nrow(df_batch_2)))
[1] "Batch 2 rows: 2"
> print(paste("Total Combined rows:", nrow(combined_data)))
[1] "Total Combined rows: 7"
> print("--- Preview (Bottom rows show the new batch) ---")
[1] "--- Preview (Bottom rows show the new batch) ---"
> print(tail(combined_data))
      id diagnosis radius_mean texture_mean
2  842517      M      20.57      17.77
3  84300903     M      19.69      21.25
4  84348301     M      11.42      20.38
5  84358402     M      20.29      14.34
6   999001     M      15.50      20.10
7   999002     B      12.30      18.50
```